



ChengetAI

Check it. Report it safely. Know who to call next.

Scam protection for African mobile money — detection that ends in help, not just a verdict.

"Chengeta" — Shona: to protect, to guard

OSF × Andela Hackathon · Safety, Reporting & Protection

chengetai.vercel.app

THE PROBLEM

Mobile money is Africa's payment rail — and its primary fraud surface.

Losses land hardest on the poorest, exactly where financial inclusion is deepening fastest. Four gaps make it worse than it needs to be.

1

The scam crosses borders. The warning doesn't.

A number burned in Lagos works fine in Accra the next day — nothing connects the two.

2

Generic filters don't know the local scripts.

Western-trained spam filters miss every local wallet's terminology and the scams built on it.

3

Reporting goes nowhere.

People are told to report it — not to whom, in what order, or how fast. The money is gone by then.

4

Reporting costs safety.

Demanding an identity silences exactly the people most exposed to retaliation.

One script. Every wallet. Seven borders.

“ *Confirmed. You have received \$80. This was sent to you in error — please reverse the money to 077... and we will not report it.*

The wrong-deposit scam — the single most reported pattern across every market we cover.

Runs identically on:

EcoCash

M-PESA

MTN MoMo

Airtel Money

OPay

Capitec Pay

So the defence has to cross borders too.

One shared scam-number database across all seven markets. A number reported in Lagos is visible to someone in Accra — and a number reported from two countries goes to high risk regardless of volume, because cross-border reach means an organised operation rather than a local dispute.

THE SOLUTION

Detect → report safely → reach help.

Most scam tools stop at the first beat. The second and third are what the Safety, Reporting & Protection track actually asks for.

01

Check it

Paste a suspicious message. Verdict, the exact phrases that triggered it, and a plain-language reason — grounded in your own market's wallets and currency.

02

Report it safely

Anonymous by default: no account, no name, and the report still counts. Phone numbers, one-time codes and ID numbers are stripped before it is ever stored.

03

Reach help

What to do in the next five minutes, then who to contact in your country — wallet provider first, because it is the only one that can still stop the transfer.

A verdict without a next step is only half an answer — so every non-safe verdict ships with the pathway attached, not behind another tap.

And on Android the first beat no longer waits to be asked — Call Guard screens incoming calls, texts and WhatsApp calls as they arrive.

WHAT'S BUILT

Seven modules, one backend, one shared database.

Check Message

AI classifier, grounded per market. Verdict, confidence, risk phrases, plain-language reason.

Get Help

The ordered escalation ladder per country: wallet → regulator → police → support.

Report

Anonymous by default. Excerpts redacted at intake, before anything is stored.

Number Lookup

Crowd-sourced reputation, including which countries a number has been reported from.

Trending & Hotspots

Live weighted-rules ranking, a regional map and a cross-country rollup.

Agent Fraud Sentinel

B2B: upload an agent transaction log, get flagged anomalies with readable reasons.

Call Guard

Screens incoming calls, SMS and WhatsApp calls — natively on Android, so it works with the app closed.

Plus a public landing page and a moderation dashboard. 8 scam categories, named by mechanism rather than by wallet brand — so one taxonomy holds in every market.

It warns you before you pick up.

Pasting a message assumes the user already suspects something. Call Guard drops that assumption: the check runs as the call, text or WhatsApp call arrives.

1

Incoming call

A call-screening service resolves the caller against the shared reputation database while the phone is still ringing.

2

SMS

Sender looked up and the body classified — two independent signals, since a new number has no reputation but still sends the known script.

3

WhatsApp call

A notification listener catches VoIP calls that never reach the dialler — increasingly where the traffic actually is.

Works with the app closed

The pipeline is native Kotlin, not Flutter — a call can arrive when the engine was never started since boot. A warning that only fires while the app happens to be open is not protection.

One pipeline, one threshold

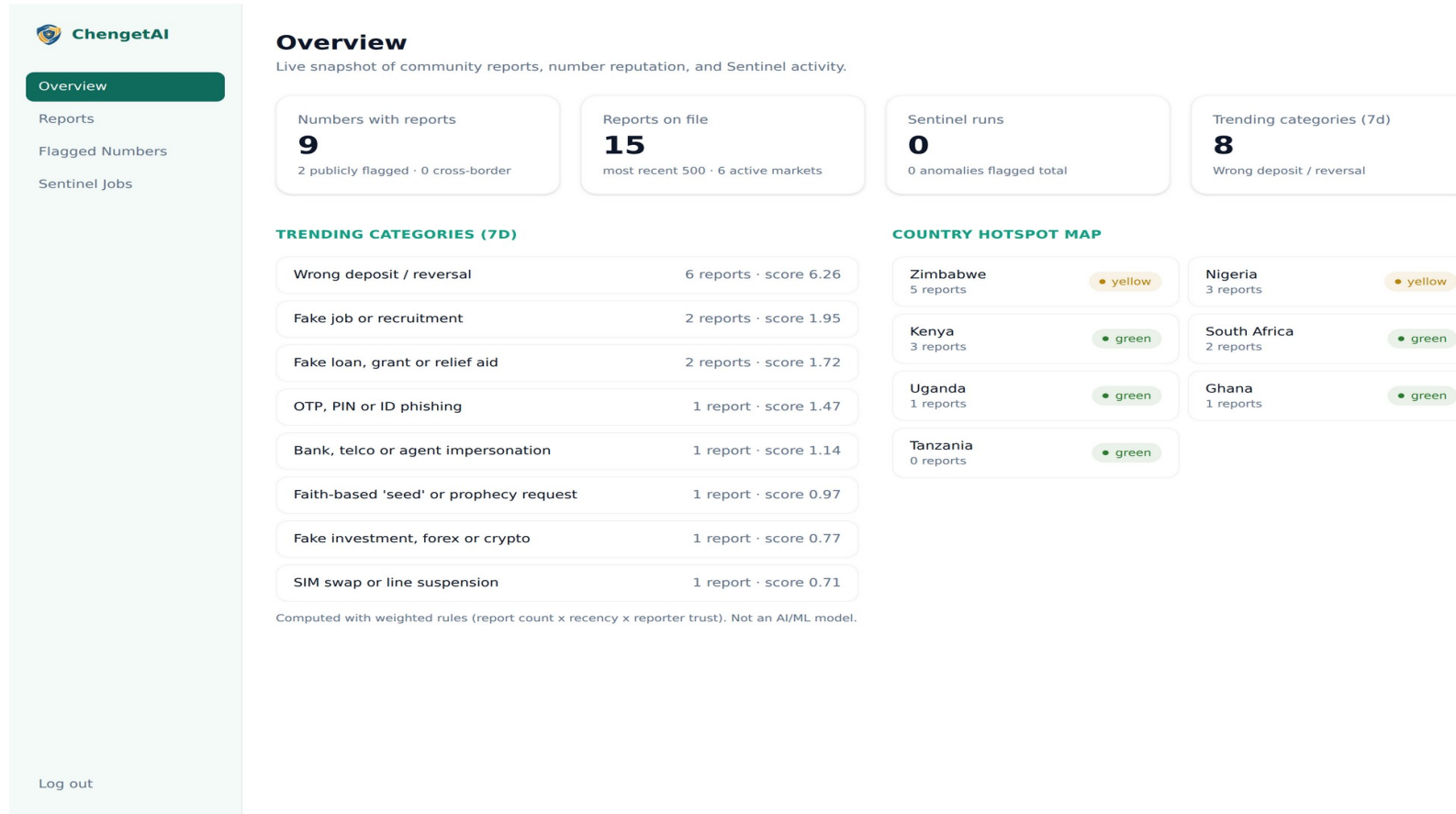
All three channels converge on the same lookup and the same warning bar, and the in-app test button drives that exact path rather than a parallel one.

Android only, and we say why

iOS forbids a live lookup on an incoming number; the nearest equivalent uploads a static blocklist in advance. That is a different feature, so we state the gap instead of shipping a weaker port.

Same database, same verdict — reaching the user through the channel the scam actually arrives on.

Not slideware — this is the running product.



Live data

Trending categories and the country hotspot map are computed from the reports table on every request.

Cross-border

Numbers reported from more than one market are flagged for moderators as an operation, not a dispute.

Stated method

The feed carries a `method_note` saying it is weighted rules, not a model — shown in the UI verbatim.

One backend. Two clients. Zero setup to evaluate.

Clients

Flutter app (Android-first) + native Kotlin call/SMS/WhatsApp screening
Next.js landing page + admin dashboard



FastAPI backend

classifier · reputation · feed · support
countries · taxonomy · redaction · sentinel



Storage

In-memory store by default
Supabase Postgres + RLS with one env var

Runs with no external services

The full API surface works on the in-memory store — no Supabase project, no API key. A reviewer clones and runs.

Adding a market is a data change

Two files: the country registry and the support-channel table. No service logic, no model, no client release.

Same code ships to production

USE_SUPABASE=true swaps the store behind a protocol. Nothing above the persistence layer changes.

We justify AI case by case — and say so at runtime.

AI

Where AI earns its place

- Message classification — scam text is adversarial and mutates constantly; a keyword blocklist catches yesterday's scripts, not tomorrow's. We ship a trained baseline and an LLM strategy so the trade-off is demonstrable, not asserted.
- Sentinel anomaly detection — agent fraud is unlabelled and drifts, which is what unsupervised learning is for. Paired with explicit rules as an honest baseline.



Where it deliberately isn't

- Number reputation and the trending feed — plain CRUD and weighted rules. The API returns a `method_note` field saying so, and the UI shows it verbatim.
- Support pathways — a curated registry, never generated. A hallucinated police hotline is worse than no hotline, so unverified contacts are shown to the user as unverified rather than hidden.
- Evidence redaction — deterministic regex, so what is removed from a stored excerpt is auditable and testable rather than a model's judgement call.

One codebase, 7 markets.

Zimbabwe

Kenya

Nigeria

Uganda

South Africa

Ghana

Tanzania

Numbers stored in E.164

0771234567 is a valid Zimbabwean, Ugandan and Tanzanian number. A national-format key would have silently merged three unrelated people's reputations — invisible in a demo, catastrophic in production.

One classifier, localised grounding

The mechanism generalises even when the vocabulary doesn't, so one model trained across all markets sees more of each pattern than seven per-country models would.

Coarse regions, two hotspot maps

A 47-county map splits reports too thin to mean anything. Cross-country rollup always; the regional breakdown only when a country is named.

TRACTION

What already exists, not what's promised.

7

markets live

behind one country registry

94

passing tests

run in CI on every push

647

labelled messages

332 scam, spread evenly
across markets

6

API modules

classify · support · reputation
· feed · reference · sentinel

Also shipped

- FastAPI backend with a full API surface that runs with zero external services
- Flutter Android-first app: check, lookup, calls, help, alerts, sentinel
- Native Android screening for incoming calls, SMS and WhatsApp calls — runs with the app closed
- Deployed landing page and a password-gated moderation dashboard
- Supabase schema with Row Level Security, and a synthetic dataset disclosed in full
- Version-pinned dependencies, CI running pytest plus web build and lint on every push

Who pays, and what happens next.

Free

Consumers

Checking, reporting, lookup and the support pathway stay free. This is the part that protects people and the part least able to pay for itself.

B2B

Agents & SMEs

Agent Fraud Sentinel licensed per agent or per till to mobile-money agent networks and SMEs who have no fraud tooling today.

Institutional

Telcos, banks, regulators

Licensed access to aggregated, anonymised cross-border scam signal — the asset no single-country tool can offer.

Path to pilot — sequenced by market, not launched everywhere at once

Now

Harden the MVP

Month 1–3

One named partner in one launch market; that market's support contacts verified; dispute flow built

Month 4–6

On-device classifier, multi-agent Sentinel dashboards, second market

Month 7–12

Telco, regulator and bank partnerships; iOS; per-jurisdiction data-protection audit

What we are not claiming.

Every one of these is disclosed in the repository rather than left for a reviewer to find.



Support contacts are mostly unverified

Only long-standing national short codes are marked verified. The rest name the right organisation with an unconfirmed number — surfaced to the user as unverified, and blocking before a pilot in that market.



The product is English-only

Interface, corpus, prompt and explanations. Across these markets the people most exposed to mobile-money fraud overlap heavily with those least likely to read English comfortably.



The data is synthetic

647 generated messages and 200 generated transactions, fully disclosed. No real user messages or transactions were collected.



No per-jurisdiction privacy review

Seven overlapping regimes. The design targets their common floor; the country-by-country audit is required before real user data and is not done.



No pilot partner yet

A target profile, not a secured partnership. Stated as pending rather than filled with a name.

The ask

People can only act on information they can verify.

ChengetAI makes a suspicious message checkable, a report safe to file, and help findable — in that order.

1

A pilot partner

An introduction to one agent network or telco district in a single launch market.

2

Contact verification

Help confirming national support-desk numbers — the blocker before any pilot.

3

Mentorship

On the dispute and human-review flow, and on what non-English coverage really takes.

Built by

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